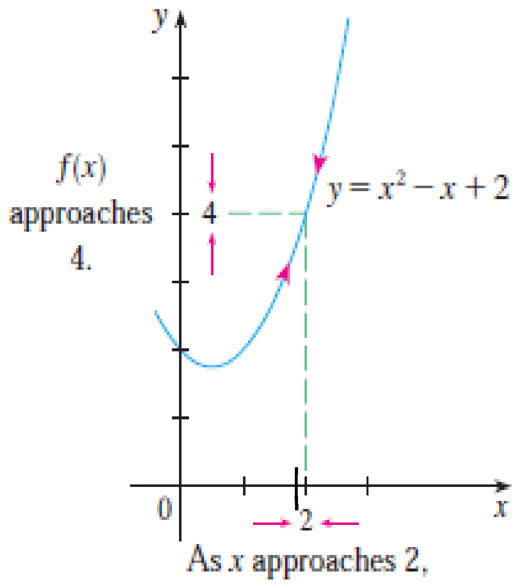
Unit 2.2 The limit of a function p83

Note Title 2014/02/19

Say $f(x) = x^2 - x + 2$ Textbook example



x	$f(x)$	x	$f(x)$
1.0	2.000000	3.0	8.000000
1.5	2.750000	2.5	5.750000
1.8	3.440000	2.2	4.640000
1.9	3.710000	2.1	4.310000
1.95	3.852500	2.05	4.152500
1.99	3.970100	2.01	4.030100
1.995	3.985025	2.005	4.015025
1.999	3.997001	2.001	4.003001

Definition p83

$f(x)$ is defined near the number a, then

$$\lim_{x \to a} f(x) = L$$

"the limit of $f(x)$, as x approaches a, equals L".

We can make values of $f(x)$ arbitrarily close to L by taking x sufficiently close to a, but not equal to a.

Example: $f(x) = \frac{e^x - 1}{x}$

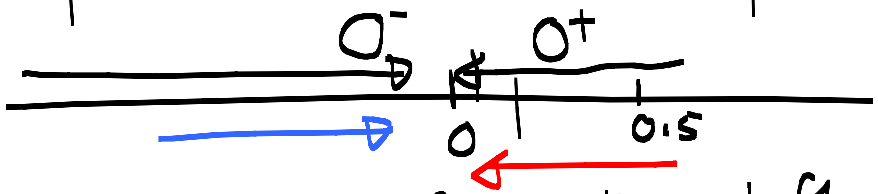
$f(1) = \frac{e-1}{1}$ $f(0) = ?$

$f(0)$ is not defined since $0 \notin D_f$

<u>However</u>:

x	$f(x)$
0.5	1.297
0.2	1.107
0.1	1.052

x	$f(x)$
−0.5	0.790
−0.2	0.906
−0.1	0.952



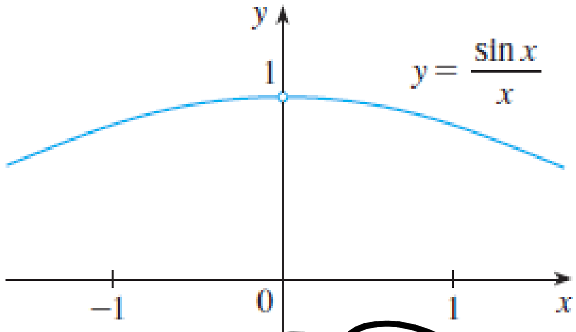
As x approaches 0 from the left is "seems" as if $f(x)$ goes to 1 and as x approaches 0 from the right it "seems" as if $f(x)$ goes to 1.

By the definition we say:

$$\lim_{x \to 0} \frac{e^x - 1}{x} = 1$$ although $f(0)$ does not exist.

A good one to remember.

x	$\frac{\sin x}{x}$
± 1.0	0.84147098
± 0.5	0.95885108
± 0.4	0.97354586
± 0.3	0.98506736
± 0.2	0.99334665
± 0.1	0.99833417
± 0.05	0.99958339
± 0.01	0.99998333
± 0.005	0.99999583
± 0.001	0.99999983



$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

We cannot make a table of values everytime we need to find a limit so

How can we calculate a limit?

How can we calculate a limit?

1. Direct substitution.

(i) $\lim_{x \to 2} \frac{x^2-4}{x+2} = \frac{0}{4} = 0$

(ii) $\lim_{x \to \pi} \cos x = -1$

2. Factorization.

(i) $\lim_{x \to 2} \frac{x^2-4}{x-2}$

$= \lim_{x \to 2} \frac{\cancel{(x-2)}(x+2)}{\cancel{x-2}} = 4$

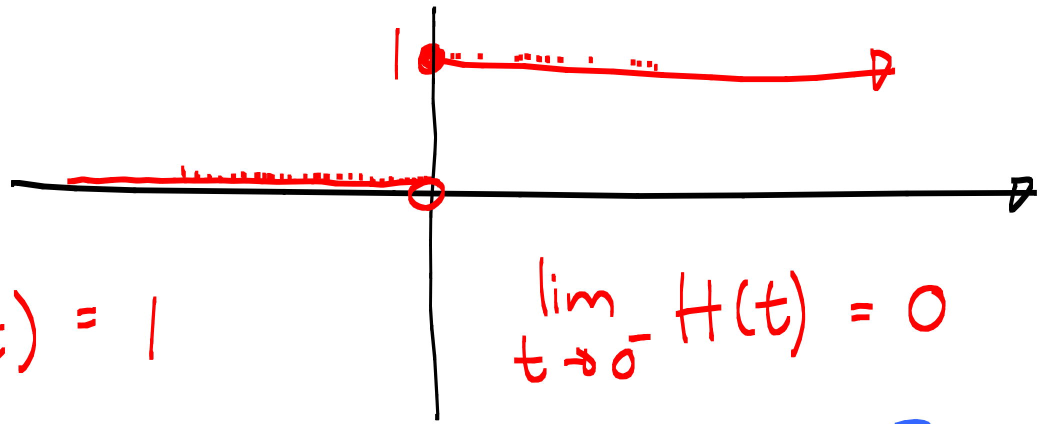
(ii) $\lim_{x \to 3} \frac{x^3-27}{x-3} = \lim_{x \to 3} \frac{\cancel{(x-3)}(x^2+3x+9)}{\cancel{x-3}}$

$= 27$

One sided limits p91

(3) $H(t) = \begin{cases} 0 & \text{if } t<0 \\ 1 & \text{if } t \geq 0 \end{cases}$

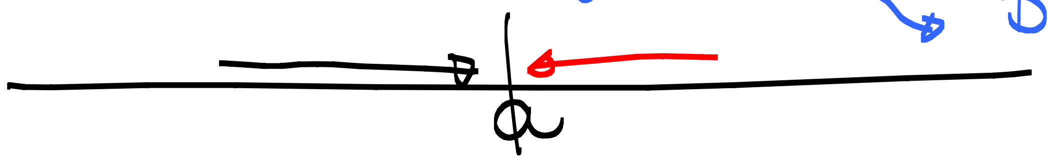
Heavyside:



$\lim_{t\to 0^+} H(t) = 1$ $\qquad$ $\lim_{t\to 0^-} H(t) = 0$

In general:

$\lim_{t\to 0} H(t) = ?$ $\to$ DNE



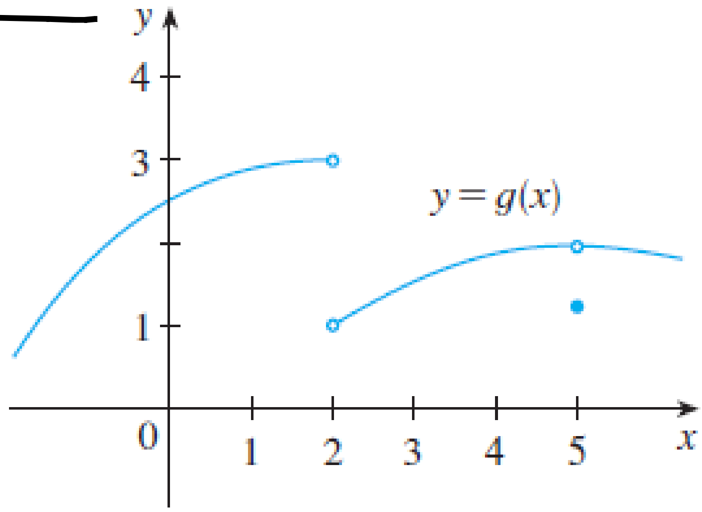
$\lim_{x\to a^-} f(x)$ left hand limit

$\lim_{x\to a^+} f(x)$ right hand limit

Now $\lim_{x\to a^-} f(x) = L$ and $\lim_{x\to a^+} f(x) = L$

$\iff \lim_{x\to a} f(x) = L$

Example 7 p 88



State the following if they exist:

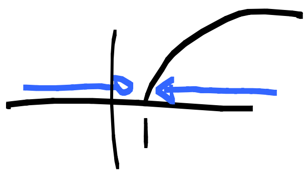
(a) $\lim_{x \to 2^-} g(x) = 3$ (b) $\lim_{x \to 2^+} g(x) = 1$

(c) $\lim_{x \to 2} g(x)$ DNE (d) $\lim_{x \to 5^-} g(x) = 2$

(e) $\lim_{x \to 5^+} g(x) = 2$ (f) $\lim_{x \to 5} g(x) = 2$

(g) $g(5) = 1$

More examples:



1. $\lim_{x \to 1^+} \sqrt{x-1}$

$= 0$

$\lim_{x \to 1^-} \sqrt{x-1}$ $x-1 \geq 0$

$x \geq 1$

DNE.

$\lim_{x \to 1} \sqrt{x-1}$ DNE.

2. $f(x) = \begin{cases} 2x+1 & \text{if} \quad x < -2 \\ x^2 - 1 & \text{if} \quad -2 < x < 1 \\ 1 - x & \text{if} \quad x > 1 \end{cases}$

$\lim_{x \to 0} f(x) = -1$

Direct.

$\lim_{x \to 1} f(x) = 0$ because.

(i) $\lim_{x \to 1^-} f(x) = 0$ (ii) $\lim_{x \to 1^+} f(x) = 0$

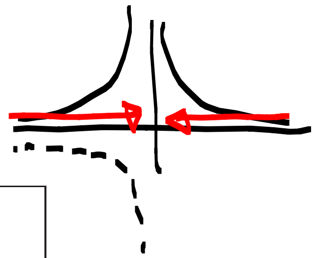
$\lim_{x \to -2} f(x)$ DNE.

(i) $\lim_{x \to -2^-} f(x) = -3$ (ii) $\lim_{x \to -2^+} f(x) = 3$

Unit 2.2 Continued

Infinite Limits

Example 8 p 93



Find $\lim_{x \to 0} \frac{1}{x^2}$

$= ?$

x	$\frac{1}{x^2}$
± 1	1
± 0.5	4
± 0.2	25
± 0.1	100
± 0.05	400
± 0.01	10,000
± 0.001	1,000,000

This behaviour is reflected with the notation:

limit DNE.

$$\lim_{x \to 0} \frac{1}{x^2} = \infty$$

not a number.

$2 \times \infty$ $\quad \infty - \infty$ $\quad \frac{\infty}{\infty}$

* See definition 4, 5 on p 89 and 90.

Let's look at: c.

$$\lim_{x \to 1} \frac{1}{(x-1)^2} = \infty$$

denominator becomes very small.

$$\frac{\text{Non zero number}}{\text{number getting smaller and smaller}} = \infty$$

* Sign!!

Examples:

$$\lim_{x \to 1^+} \frac{-4 \ \ominus}{x-1 \ \oplus} = -\infty$$

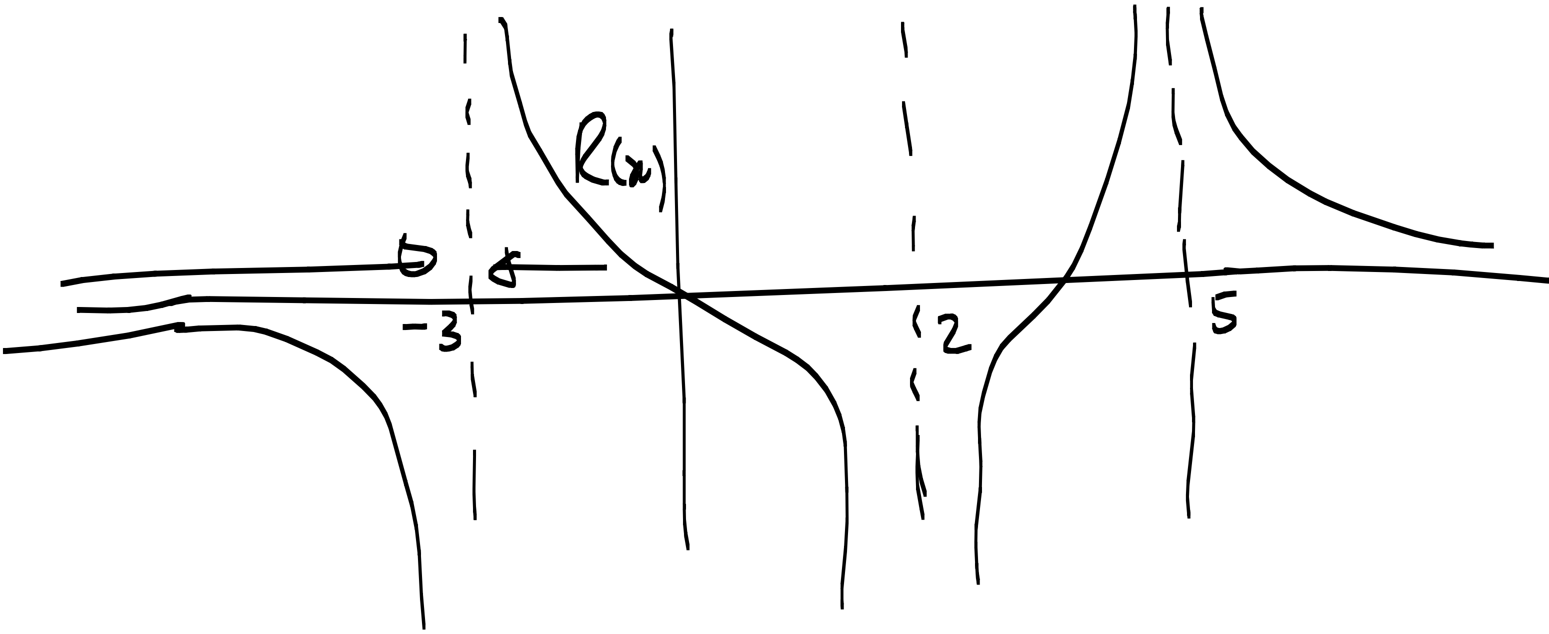
$$\lim_{x \to 1^-} \frac{-4 \ \ominus}{x-1 \ \ominus} = +\infty$$

① Sketch a graph that satisfies the following:

$$\lim_{x \to 1^-} f(x) = 3 \qquad \lim_{x \to 1^+} f(x) = 7$$

$f(1) = 2$

②



(a) $\lim_{x \to 2} R(x) = -\infty$ (b) $\lim_{x \to 5} R(x) = +\infty$

(c) $\lim_{x \to -3^-} R(x) = -\infty$ (d) $\lim_{x \to -3^+} R(x) = +\infty$

Vertical asymptotes

The line $x = a$ is called a vertical asymptote of the curve $y = f(x)$ if at least one of the following is true:

$$\lim_{x \to a} f(x) = \infty, \quad \lim_{x \to a} f(x) = -\infty$$

$$\lim_{x \to a^-} f(x) = \infty, \quad \lim_{x \to a^-} f(x) = -\infty$$

$$\lim_{x \to a^+} f(x) = \infty, \quad \lim_{x \to a^+} f(x) = -\infty$$

Example: $f(x) = \frac{1}{x^2}$ has a vertical asymptote $x = 0$ since

$$\lim_{x \to 0} \frac{1}{x^2} = \infty$$

How do we find the vertical asymptotes?

Find the domain:

- If a is in the domain then $x = a$ is not a vertical asympt.

- If a is not in the domain, you must <u>test</u> whether $x = a$ is a vertical asymptote.

<u>Example:</u>

Find the vertical asymptote (if it exists) for $f(x) = \frac{-4}{x-1}$ $\quad D_f = \mathbb{R} \setminus \{1\}$

So $x = 1$ is a possible V.A.

$$\lim_{x \to 1^-} \frac{-4}{x-1} = \underline{+\infty} \qquad \underbrace{\lim_{x \to 1^+} \frac{-4}{x-1} = -\infty}_{\text{not necessary.}}$$

If <u>one</u> of them goes to ∞ then $x = 1$ is a V.A.

<u>More examples:</u>

Find the vertical asymptotes, if any:

(1) $f(x) = x + 1$ $\quad D_f = \mathbb{R} \Rightarrow$ No V.A.

(2) $f(x) = \dfrac{\sqrt{x}-3}{x-9}$

$D_f = [0,9) \cup (9,\infty)$

Possible V.A at $x=0$ and $x=9$

Test: $\displaystyle\lim_{x\to 0^+} \frac{\sqrt{x}-3}{x-9} = \frac{1}{3}$ (Direct subst)

$\Rightarrow$ $x=0$ is not a V.A.

$\displaystyle\lim_{x\to 9^+} \frac{\cancel{\sqrt{x}-3}}{(\cancel{\sqrt{x}-3})(\sqrt{x}+3)} = \frac{1}{6}$ and

$\displaystyle\lim_{x\to 9^-} \frac{\cancel{\sqrt{x}-3}}{(\cancel{\sqrt{x}-3})(\sqrt{x}+3)} = \frac{1}{6}$

$\Rightarrow$ $x=9$ is not a V.A.

(3) $f(x) = \dfrac{1-x}{x-5}$ $\qquad D_f = \mathbb{R} \setminus \{5\}$

So $x=5$ is a possible V.A.

$$\left.\begin{aligned} \lim_{x \to 5^-} \frac{1-x}{x-5} \begin{matrix}\ominus\\ \ominus\end{matrix} &= \infty \\ \lim_{x \to 5^+} \frac{1-x}{x-5} \begin{matrix}\ominus\\ \oplus\end{matrix} &= -\infty \end{aligned}\right\} \Rightarrow x=5 \text{ is}$$

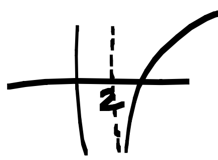
(Sign!)

(4) $f(x) = \ln(\underbrace{x-2}_{})$ $\qquad D_f = (2, \infty)$

$\to x-2>0$

$x>2$

A possible VA at $x=2$



$$\lim_{x \to 2^+} \ln(x-2) = -\infty$$

$\Rightarrow x=2$ is a V.A.

(5) $\displaystyle \lim_{x \to 2^-} \frac{6}{(\underset{-}{x-2})(\underset{+}{7-x})} \begin{matrix}\oplus\\ \ominus\end{matrix}$

$\dfrac{c}{0}$ $= -\infty$

Sign

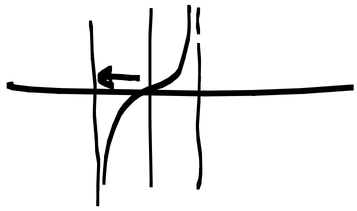
(6) $\displaystyle \lim_{x \to 4^+} \frac{x^2 - x}{(x-4)^2} \begin{matrix}\oplus\\ \oplus\end{matrix}$

$= +\infty$

(6) V.A of $f(x) = \ln x$.

$\displaystyle \lim_{x \to 0^+} f(x) = -\infty$ $\quad \therefore x=0$ is VA.

(7) V.A of $f(x) = \tan x$.



$$\lim_{x \to \frac{\pi}{2}^-} \tan x = +\infty$$

$x = \frac{\pi}{2}$ is a V.A.

$$\lim_{x \to -\frac{\pi}{2}^+} \tan x = -\infty$$

$x = -\frac{\pi}{2}$ is a V.A.

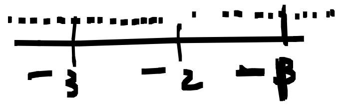
$$x = \frac{\pi}{2}(2n+1), \quad n \in \mathbb{Z}.$$

Find the V.A. $f(x) = \dfrac{x-1}{x^2(x+2)}$

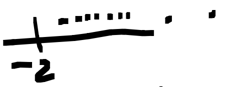
and sketch(?) it.

$D_f = \mathbb{R} \setminus \{0, -2\}$ Possible V.A. $x=0$, $x=-2$

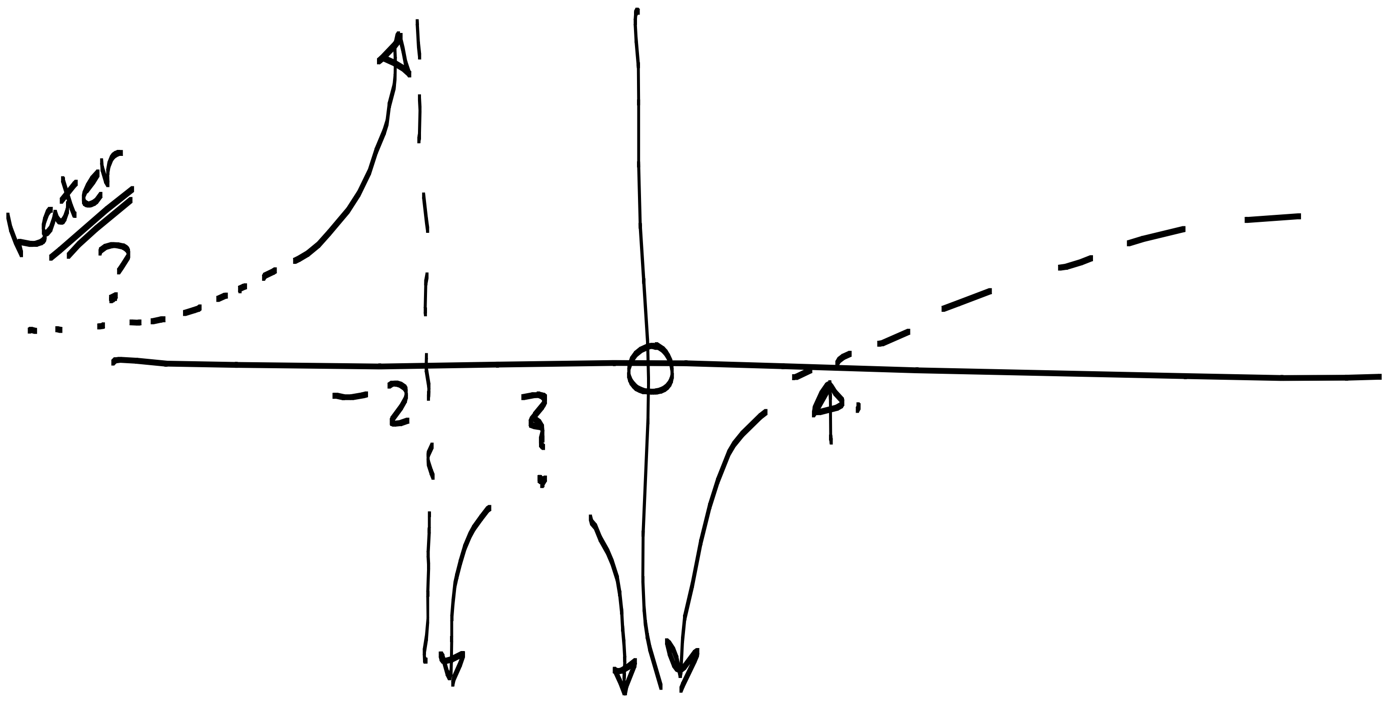
$$\lim_{x \to 0^+} \frac{x-1}{x^2(x+2)} = -\infty \qquad \lim_{x \to 0^-} \frac{x-1}{x^2(x+2)} = -\infty$$



$\therefore$ $x = 0$ is a V.A.



$$\lim_{x \to -2^+} \frac{x-1 \ \ominus}{\oplus x^2(x+2)\oplus} = -\infty \qquad \lim_{x \to -2^-} \frac{x-1 \ \ominus}{\oplus x^2(x+2)\ \ominus} = +\infty$$


Later
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4.